Table 1: Privacy parameters and model utility for each testbed and noise level. DP-SGD pipeline calibrated for per-sample gradient compatibility: WRN BatchNorm replaced with GroupNorm; larger batch sizes (e.g. 1024 on CIFAR-10/100) and retuned learning rates to improve gradient SNR under noise injection. Image testbeds suffer a substantial accuracy drop vs. the main paper (-12.7 pp on CIFAR-10, -27.5 pp on CIFAR-100), driven by the BatchNorm→GroupNorm switch; tabular testbeds (Location, Purchase) use plain MLPs without normalization layers and benefit modestly from the retuning. *All baselines and defended models share this calibration*, so the only variable across rows within a testbed is the noise multiplier σ. Absolute risk metrics in this document are therefore not directly comparable to the main paper; interpret them *relative* across noise levels within this document. δ is set to $\delta \approx 1/n_{\text{train}}$; $C = 10$ for all DP-SGD runs; ε from the Opacus accountant, rounded down to the nearest order of magnitude when > 100. At $\sigma = 0.01$ the privacy budget is vacuous ($\varepsilon > 100$K); at $\sigma = 1.0$ the guarantee becomes meaningful ($\varepsilon \approx 6$–9) but utility degrades substantially, especially for CIFAR-100 and Location.

Testbed	Defence	Test Acc (%)	ε	δ
CIFAR-10 / WRN	Baseline	76.1	—	—
	$\sigma = 0.01$	71.6	>1M	3.3×10^{-5}
	$\sigma = 0.1$	71.2	>10K	3.3×10^{-5}
	$\sigma = 1.0$	54.1	6.7	3.3×10^{-5}
CIFAR-100 / WRN	Baseline	40.0	—	—
	$\sigma = 0.01$	27.7	>1M	3.3×10^{-5}
	$\sigma = 0.1$	26.4	>10K	3.3×10^{-5}
	$\sigma = 1.0$	9.0	6.1	3.3×10^{-5}
Location / MLP3	Baseline	66.4	—	—
	$\sigma = 0.01$	64.6	>100K	4.0×10^{-4}
	$\sigma = 0.1$	55.8	>1K	4.0×10^{-4}
	$\sigma = 1.0$	7.1	8.9	4.0×10^{-4}
Purchase / MLP3	Baseline	89.6	—	—
	$\sigma = 0.01$	87.3	>1M	1.0×10^{-5}
	$\sigma = 0.1$	86.4	>1K	1.0×10^{-5}
	$\sigma = 1.0$	77.3	5.7	1.0×10^{-5}

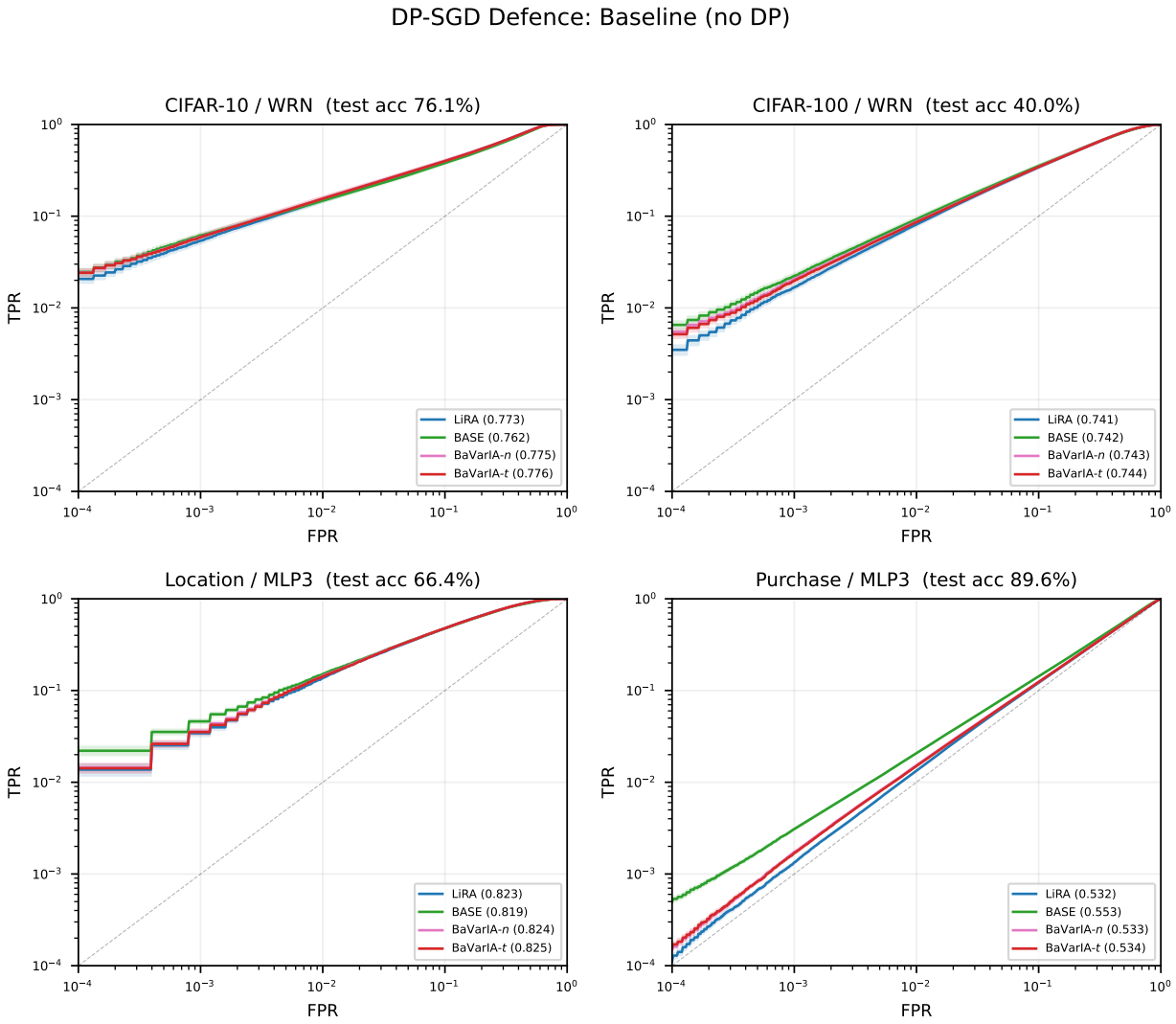

Figure 1: Defence-level view, baseline (no DP-SGD): all four membership-inference attacks (LiRA, BASE, BaVarIA-n, BaVarIA-t) overlaid on each of four testbeds (CIFAR-10/WRN, CIFAR-100/WRN, Location/MLP3, Purchase/MLP3). $K = 64$ shadow models, online setting, mean ROC over 32 replicates on log–log axes; legend AUC included. Panel titles include mean test accuracy across shadow models.

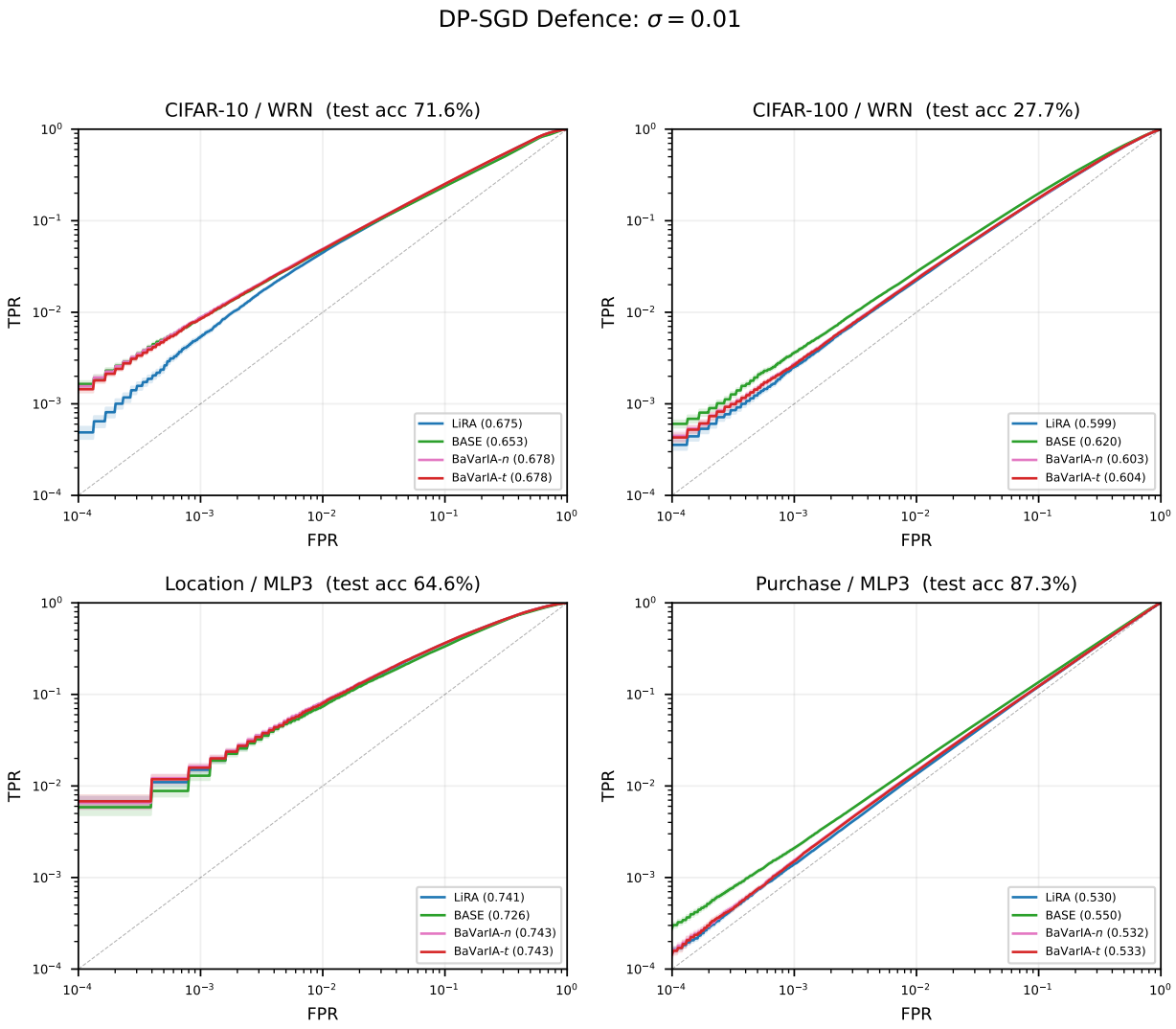

Figure 2: Defence-level view, DP-SGD with $\sigma = 0.01$, $C = 10$. LiRA, BASE, BaVarIA-n, BaVarIA-t overlaid on each testbed; $K = 64$, mean over 32 replicates. Privacy budget at this noise level is vacuous ($\varepsilon > 100$K).

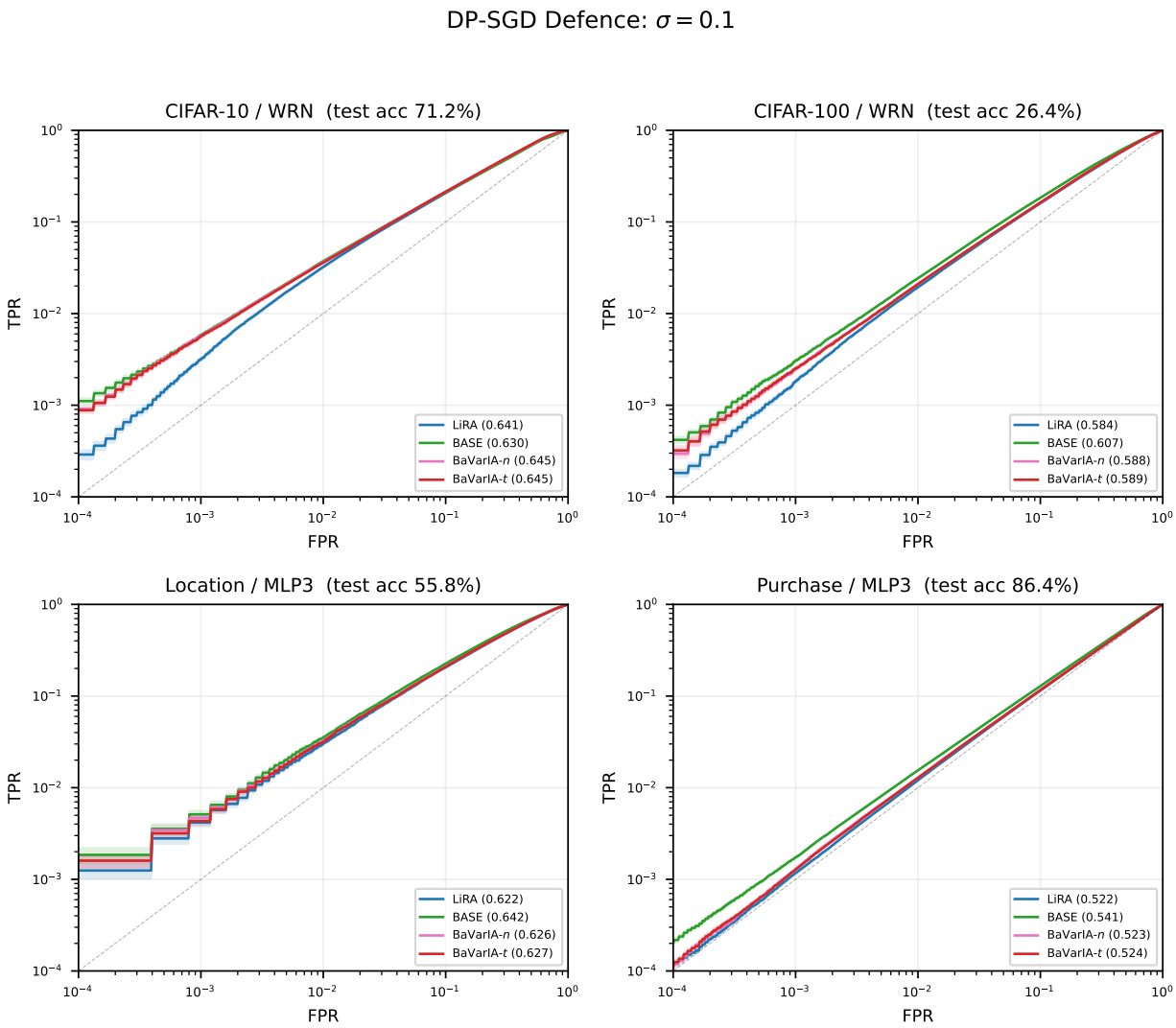

Figure 3: Defence-level view, DP-SGD with $\sigma = 0.1$, $C = 10$. LiRA, BASE, BaVarIA-n, BaVarIA-t overlaid; $K = 64$, mean over 32 replicates.

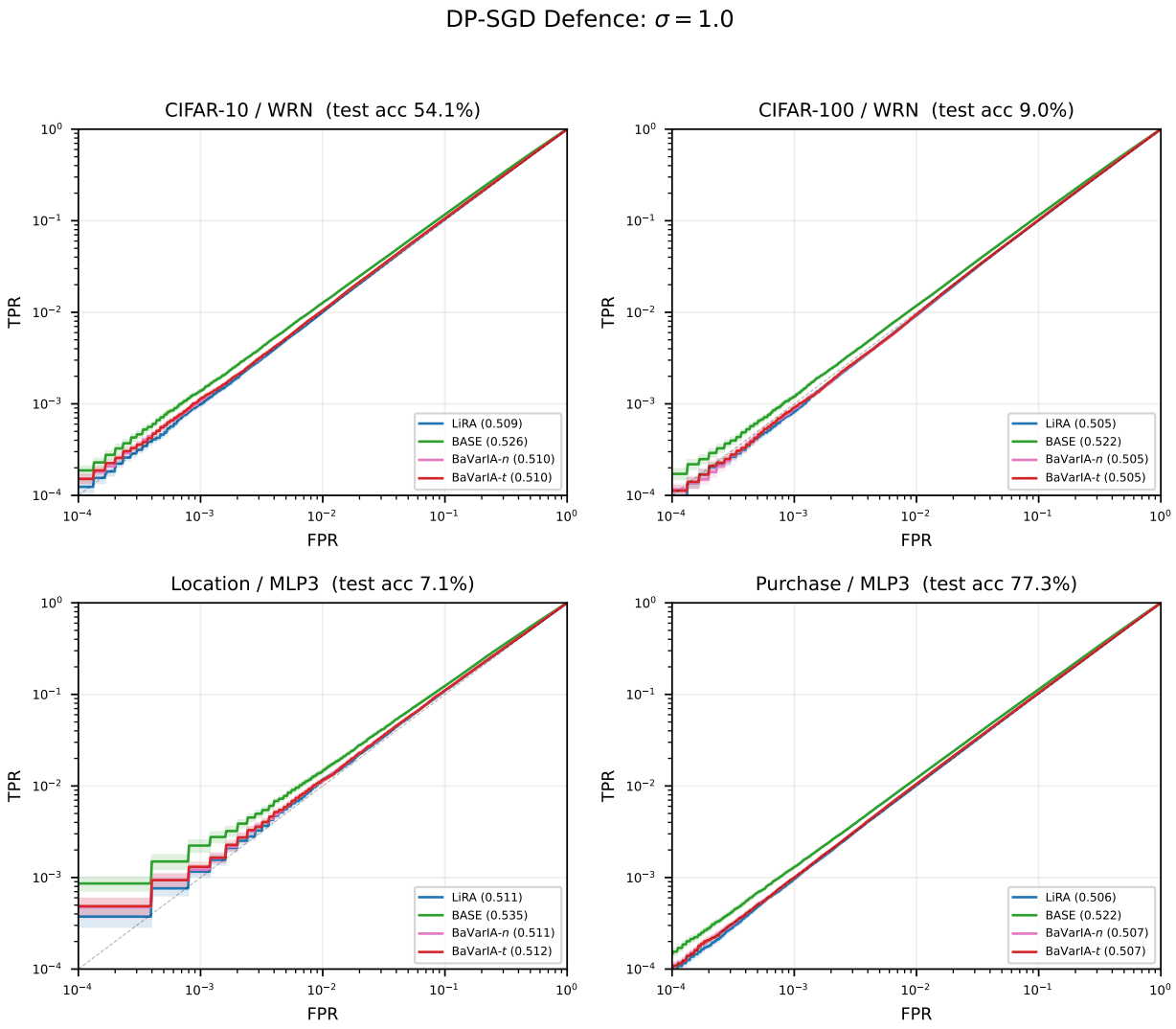

Figure 4: Defence-level view, DP-SGD with $\sigma = 1.0$, $C = 10$ (meaningful privacy, $\varepsilon \approx 6$–9). LiRA, BASE, BaVarIA-n, BaVarIA-t overlaid; $K = 64$, mean over 32 replicates. All attacks approach random guessing at this noise level.

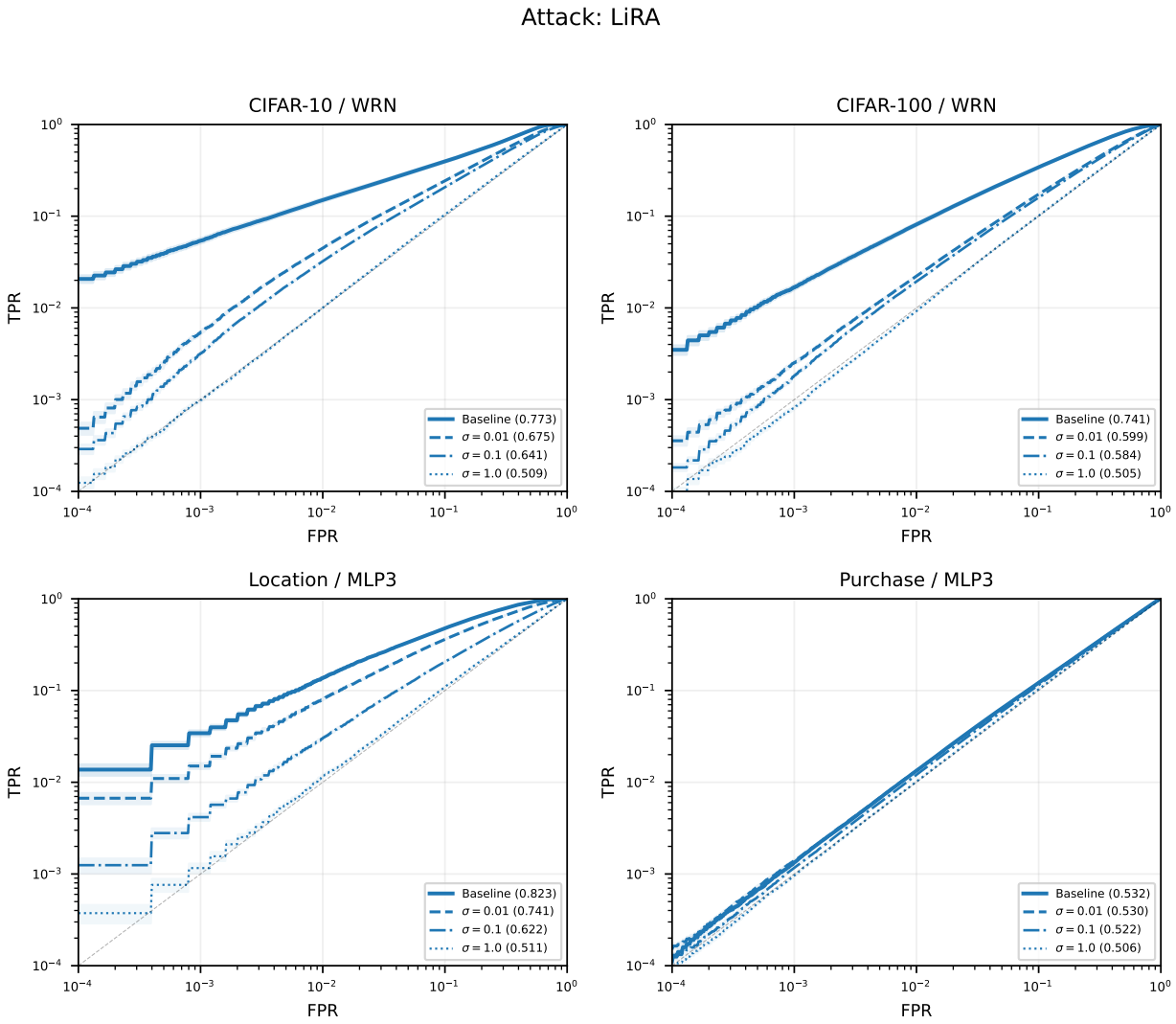

Figure 5: Attack-method view, LiRA across defence levels $\sigma \in \{0, 0.01, 0.1, 1.0\}$ on each testbed. Linestyle distinguishes defence level; $K = 64$, online, mean over 32 replicates, log–log axes.

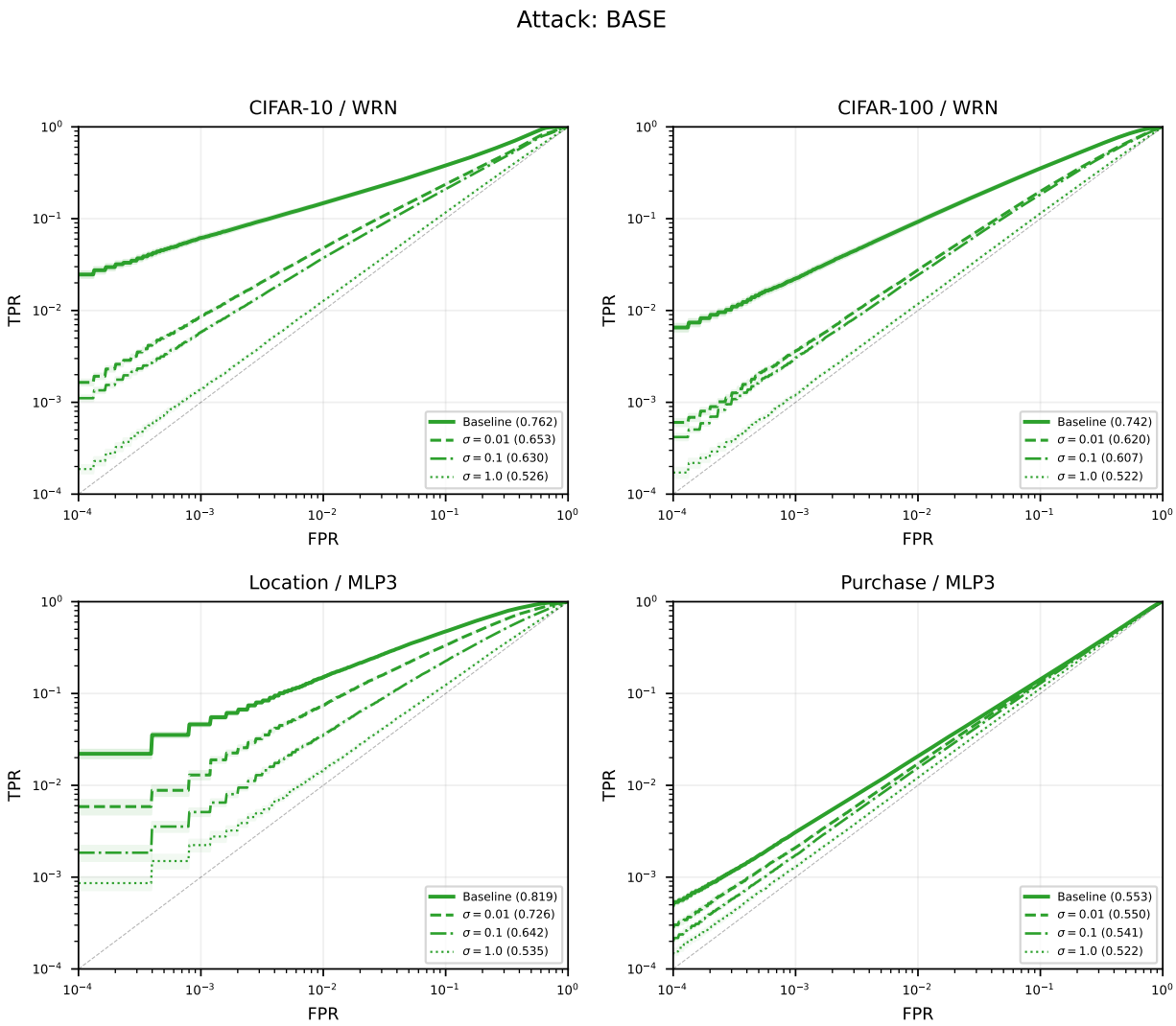

Figure 6: Attack-method view, BASE across defence levels $\sigma \in \{0, 0.01, 0.1, 1.0\}$. Linestyle distinguishes defence level; $K = 64$, online, mean over 32 replicates.

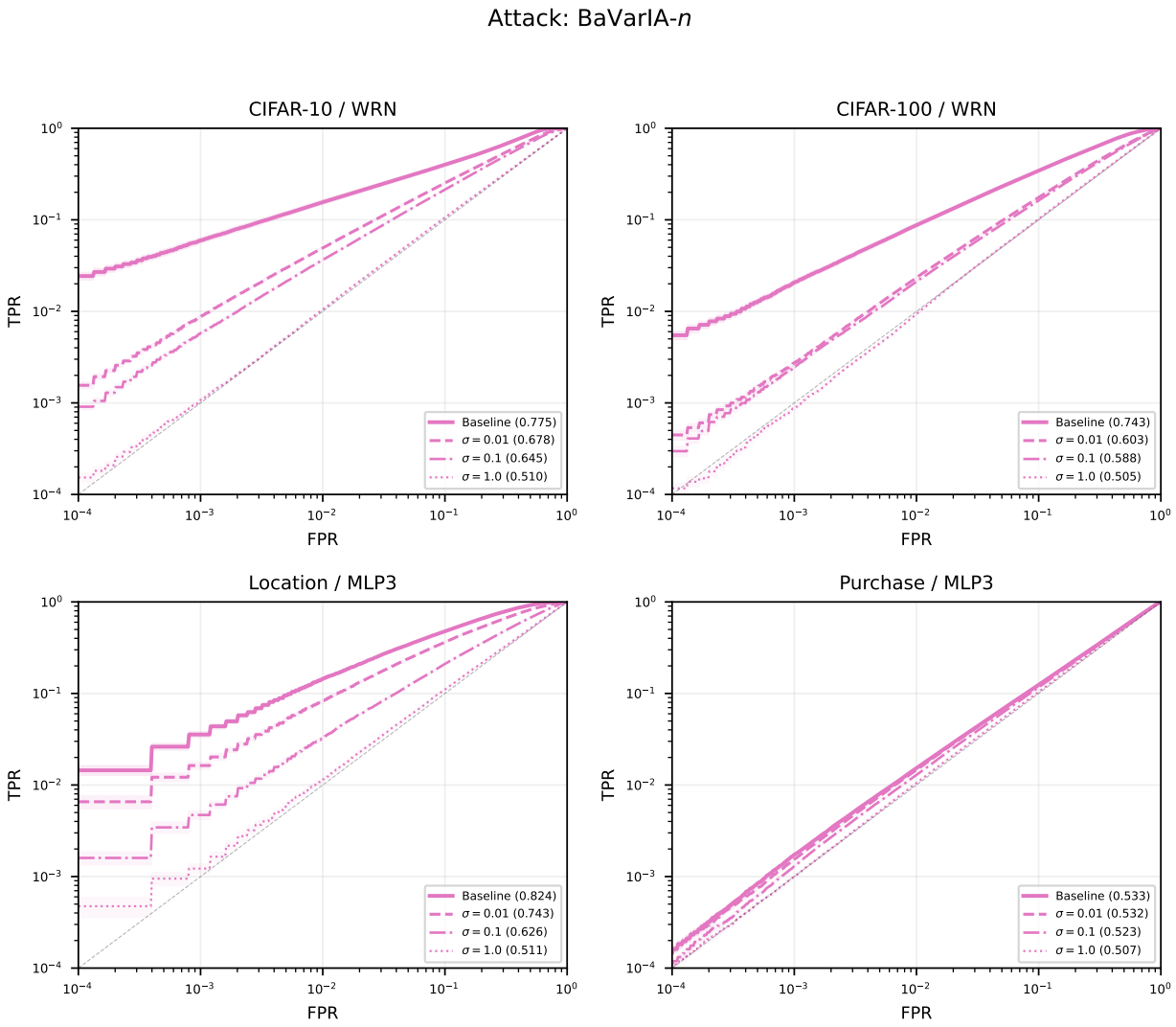

Figure 7: Attack-method view, BaVarIA-n across defence levels $\sigma \in \{0, 0.01, 0.1, 1.0\}$. Linestyle distinguishes defence level; $K = 64$, online, mean over 32 replicates.

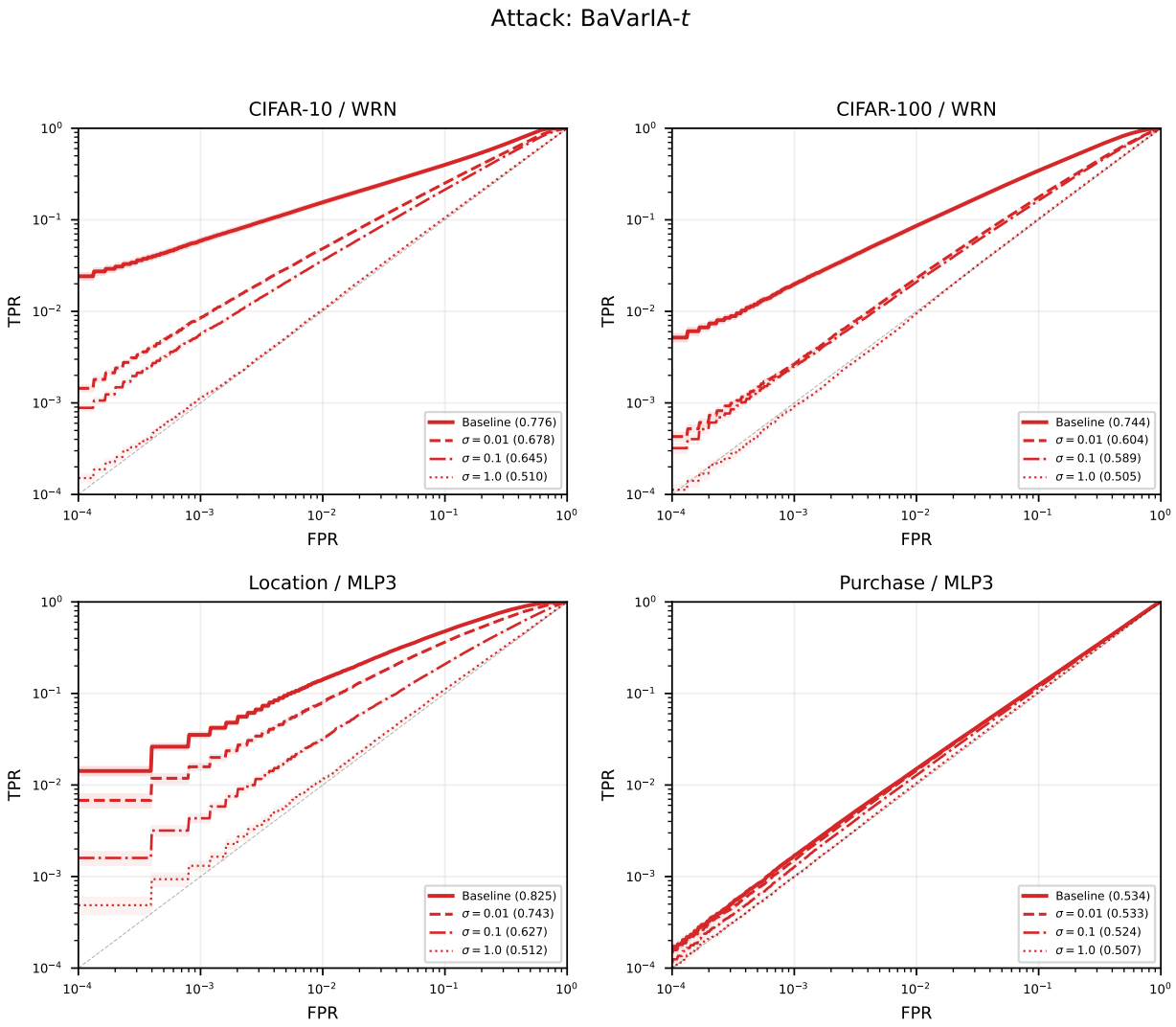

Figure 8: Attack-method view, BaVarIA-t across defence levels $\sigma \in \{0, 0.01, 0.1, 1.0\}$. Linestyle distinguishes defence level; $K = 64$, online, mean over 32 replicates.

Table 2: AUC across testbeds and defence levels. Mean over 32 replicates, $K = 64$, online setting. Best method per row in bold. BaVarIA-n / BaVarIA-t track or beat LiRA at every noise level (16/16 cells in AUC); BASE progressively overtakes the Gaussian methods as σ grows, winning 4/4 testbeds at $\sigma = 1.0$. The crossover mirrors the small-K regime in the absence of defence — simpler distributional families dominate when per-sample signal is scarce, whether scarcity comes from few shadow models or from added DP-SGD noise.

Testbed	Defence	Acc (%)	LiRA	BASE	BaVarIA-n	BaVarIA-t
CIFAR-10 WRN	Baseline	76.1	.773	.762	.775	**.776**
	$\sigma = 0.01$	71.6	.674	.653	**.678**	.678
	$\sigma = 0.1$	71.2	.641	.630	**.645**	.645
	$\sigma = 1.0$	54.1	.509	**.526**	.510	.510
CIFAR-100 WRN	Baseline	40.0	.741	.742	.743	**.744**
	$\sigma = 0.01$	27.7	.600	**.620**	.603	.604
	$\sigma = 0.1$	26.4	.584	**.607**	.588	.589
	$\sigma = 1.0$	9.0	.505	**.522**	.505	.505
Location MLP3	Baseline	66.4	.823	.819	.824	**.825**
	$\sigma = 0.01$	64.6	.741	.726	.743	**.743**
	$\sigma = 0.1$	55.8	.622	**.642**	.626	.627
	$\sigma = 1.0$	7.1	.511	**.535**	.511	.512
Purchase MLP3	Baseline	89.6	.532	**.553**	.533	.534
	$\sigma = 0.01$	87.3	.530	**.550**	.532	.533
	$\sigma = 0.1$	86.4	.522	**.541**	.523	.524
	$\sigma = 1.0$	77.3	.506	**.522**	.507	.507

Table 3: TPR at FPR= 0.01 across testbeds and defence levels. Mean over 32 replicates, $K = 64$, online setting. Best method per row in bold. The Gaussian-family BaVarIA-n holds or beats LiRA throughout; BASE leads at heavy noise. At $\sigma = 1.0$ all attacks approach random guessing (TPR$\leq$ 0.015 vs. random baseline 0.01).

Testbed	Defence	Acc (%)	LiRA	BASE	BaVarIA-n	BaVarIA-t
CIFAR-10 WRN	Baseline	76.1	.150	.148	**.156**	.156
	$\sigma = 0.01$	71.6	.045	.048	**.049**	.049
	$\sigma = 0.1$	71.2	.032	**.038**	.037	.036
	$\sigma = 1.0$	54.1	.010	**.013**	.011	.011
CIFAR-100 WRN	Baseline	40.0	.082	**.093**	.087	.086
	$\sigma = 0.01$	27.7	.022	**.028**	.023	.023
	$\sigma = 0.1$	26.4	.019	**.024**	.021	.021
	$\sigma = 1.0$	9.0	.009	**.012**	.009	.010
Location MLP3	Baseline	66.4	.139	**.153**	.146	.144
	$\sigma = 0.01$	64.6	.081	.075	**.084**	.081
	$\sigma = 0.1$	55.8	.031	**.036**	.033	.032
	$\sigma = 1.0$	7.1	.012	**.015**	.012	.012
Purchase MLP3	Baseline	89.6	.013	**.021**	.015	.015
	$\sigma = 0.01$	87.3	.013	**.017**	.015	.014
	$\sigma = 0.1$	86.4	.012	**.016**	.013	.013
	$\sigma = 1.0$	77.3	.010	**.012**	.011	.011

Table 4: TPR at FPR= 0.001 across testbeds and defence levels. Mean over 32 replicates, $K = 64$, online setting. Best method per row in bold. The intermediate regime $\sigma = 0.01$ is notable: it degrades accuracy only modestly (CIFAR-10 76%→72%, Purchase 90%→87%) while substantially reducing attack TPR, but the formal privacy guarantee at this level is vacuous ($\varepsilon > 1$M); the practical privacy may stem from gradient clipping ($C = 10$) rather than noise injection.

Testbed	Defence	Acc (%)	LiRA	BASE	BaVarIA-n	BaVarIA-t
CIFAR-10 WRN	Baseline	76.1	.054	**.062**	.060	.059
	$\sigma = 0.01$	71.6	.005	.009	**.009**	.009
	$\sigma = 0.1$	71.2	.003	**.006**	.006	.006
	$\sigma = 1.0$	54.1	.0010	**.0014**	.0011	.0012
CIFAR-100 WRN	Baseline	40.0	.017	**.022**	.021	.020
	$\sigma = 0.01$	27.7	.003	**.004**	.003	.003
	$\sigma = 0.1$	26.4	.002	**.003**	.002	.003
	$\sigma = 1.0$	9.0	.0008	**.0012**	.0009	.0009
Location MLP3	Baseline	66.4	.034	**.046**	.036	.035
	$\sigma = 0.01$	64.6	.015	.013	**.016**	.016
	$\sigma = 0.1$	55.8	.004	**.005**	.005	.004
	$\sigma = 1.0$	7.1	.0012	**.0022**	.0012	.0013
Purchase MLP3	Baseline	89.6	.0013	**.0031**	.0017	.0017
	$\sigma = 0.01$	87.3	.0014	**.0021**	.0015	.0015
	$\sigma = 0.1$	86.4	.0012	**.0017**	.0013	.0013
	$\sigma = 1.0$	77.3	.0010	**.0013**	.0010	.0010